

IMPERIAL

Jacobian-free linear stability in Nektar++

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Goal and idea

- Instead of evolving the linear operator $\mathcal{L}$, we evolve the nonlinear time stepper $\mathcal{A}_T$ and approximate $\mathcal{L}$ with finite differences.

$$\mathcal{L}\mathbf{v} \approx \frac{\mathcal{A}_T(\bar{\mathbf{q}} + \epsilon\mathbf{v}) - \mathcal{A}_T(\bar{\mathbf{q}})}{\epsilon} \quad (\text{forward difference, 1 nonlinear run per Krylov vector})$$

- Validation so far:
 1. Incompressible: channel flow
 2. Compressible: poiseuille flow (exact parallel flow solution), compared against Orr–Sommerfeld (local LST)

Implementation (DriverModifiedArnoldi)

- Input EvolutionOperator = Nonlinear.
- $\mathcal{A}_T(\bar{\mathbf{q}})$ is computed once at start-up and subtracted in every iteration, so a base flow that is not an exact discrete steady state does not pollute the difference. We have $\|\mathcal{A}_T(\bar{\mathbf{q}}) - \bar{\mathbf{q}}\|/\|\bar{\mathbf{q}}\| \sim 10^{-12}$ for incompressible, 10^{-10} – 10^{-8} for compressible.
- ϵ is taken as

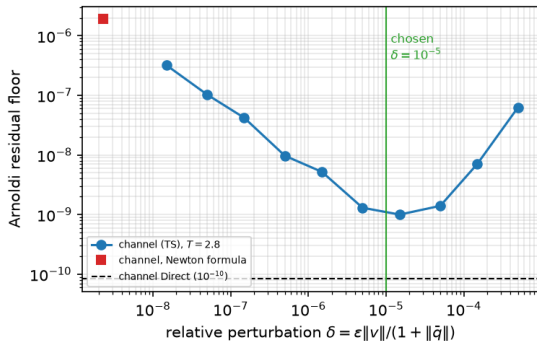
$$\epsilon\|\mathbf{v}\| = \delta(1 + \|\bar{\mathbf{q}}\|), \quad \delta = 10^{-5},$$

which differs from jacobian free Newton solver for the compressible solver:

$$\epsilon\|\mathbf{v}\| = \sqrt{\epsilon_m} \sqrt{(1 + \|\bar{\mathbf{q}}\|)}, \quad \epsilon_m = 2.22 \times 10^{-16}.$$

Choosing ϵ (or δ actually)

- Depending on the δ chosen, the Arnoldi residual may stagnate at a floor value.
- I do not use the Newton-solver formula because there Jv only steers the step and the exact residual corrects any error at the next iteration, whereas in Arnoldi the finite difference is the operator whose eigenvalues we compute, so its error is never corrected and sets the residual floor (red square: Newton formula, stalls at 2×10^{-6}).



Direct vs Nonlinear ($T = 2.8$)

Reference (ref): Direct at $\text{evtol} = 10^{-10}$, $\lambda = 0.769927 + 0.647928i \Rightarrow$ growth 2.2355×10^{-3} , freq. 0.249844.

evtol	its		$ \lambda_a - \lambda_b / \lambda_{\text{ref}} $		$\ \hat{q}_a - \hat{q}_b\ _\infty / \ \hat{q}_{\text{ref}}\ _\infty$	
	D	N	D vs N	N vs ref	D vs N	N vs ref
10^{-4}	2	2	2.7×10^{-8}	2.9×10^{-6}	4.2×10^{-5}	4.9×10^{-5}
10^{-5}	2	7	1.5×10^{-6}	1.3×10^{-6}	1.9×10^{-5}	1.9×10^{-5}
10^{-6}	10	13	3.6×10^{-7}	3.8×10^{-8}	1.5×10^{-6}	2.0×10^{-6}
10^{-7}	19	24	2.0×10^{-8}	1.3×10^{-8}	2.3×10^{-7}	2.7×10^{-7}
10^{-8}	28	37	8.0×10^{-9}	5.3×10^{-9}	3.1×10^{-8}	2.8×10^{-8}
10^{-9}	36	55	4.7×10^{-9}	4.9×10^{-9}	1.2×10^{-8}	1.2×10^{-8}
10^{-10}	45	300*	-	-	-	-

*not converged (300 its).

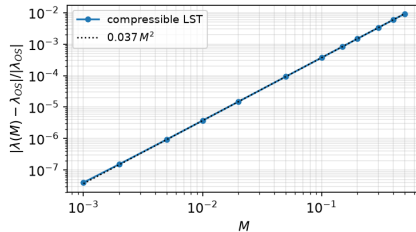
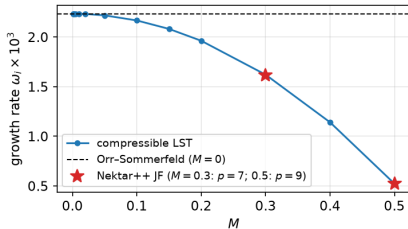
- Both converge to the same eigenvalue and eigenvector down to 10^{-9} ; Nonlinear needs $1.2\text{--}1.5 \times$ iterations.
- 10^{-10} is not reachable with Nonlinear. No matter what δ I input. Central difference $O(\delta^2)$ would solve the problem, but I don't think is preferred.

Compressible case: forced plane Poiseuille

- $Re = 7500$, $\alpha = 1$, isothermal walls ($u = v = 0$, $T_w = 1$) top and bottom, periodicity in x , constant μ and k , $Pr = 0.72$, $\gamma = 1.4$. We add a body force to the x -momentum ($f = 2\mu$) and energy equations ($\tilde{f} = fu$). In this conditions we have an exact equilibrium:

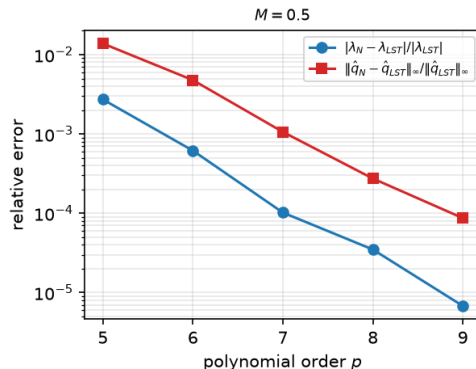
$$u = 1 - y^2, \quad v = 0, \quad T = 1 + \frac{Pr(\gamma-1)M^2}{3}(1 - y^4), \quad \rho = 1/T, \quad p = \text{const},$$

- Reference: Chebyshev local LST solver for the same equations. Tends to incompressible Orr-Sommerfeld ($\lambda_{OS} = 0.24989154 + 0.00223498i$) as for $M \rightarrow 0$:



Compressible results: Jacobian-free Arnoldi vs LST

M	p	its	$\frac{ \lambda_N - \lambda_{LST} }{ \lambda_{LST} }$	$\frac{\ \hat{\mathbf{q}}_N - \hat{\mathbf{q}}_{LST}\ _\infty}{\ \hat{\mathbf{q}}_{LST}\ _\infty}$
0.5	5	32	2.7×10^{-3}	1.4×10^{-2}
	6	31	6.2×10^{-4}	4.8×10^{-3}
	7	30	1.0×10^{-4}	1.1×10^{-3}
	8	30	3.5×10^{-5}	2.7×10^{-4}
	9	20	6.8×10^{-6}	8.7×10^{-5}
0.3	7	26	1.0×10^{-4}	1.0×10^{-3}



$\lambda = \omega_r + i \omega_i$: complex eigenvalue (frequency + growth). $\hat{\mathbf{q}} = (\rho', \mathbf{u}', \mathbf{v}', T')$. $\text{evtol} = 10^{-6}$, $T = 28$.

Compressible TS eigenfunctions

